NexaFlow S-200

Industrial Reverse-Osmosis Water Treatment Unit

Installation, Operation and Maintenance Manual

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1. Introduction and Safety

This manual describes the installation, commissioning, operation and routine maintenance of the NexaFlow S-200 industrial reverse-osmosis water treatment unit. It is intended for qualified technicians responsible for the mechanical and electrical installation of the unit, and for plant operators responsible for its day-to-day running.

The S-200 is a skid-mounted single-pass reverse-osmosis unit designed for process water production in light industrial applications such as boiler feedwater, cooling tower make-up, laboratory supply and general rinse water. It is supplied fully assembled, pre-piped and pre-wired, and requires only connection to feed water, drain, permeate outlet and mains electrical supply.

1.1 Intended use

The S-200 is designed exclusively for the treatment of municipal or pre-treated well water within the feed water limits given in Section 3. Use of the unit outside these limits, or with feed water containing free chlorine, oils, hydrocarbons or suspended solids above the stated limit, will damage the membrane elements and voids the warranty.

The unit is not designed or certified for the production of potable drinking water, for pharmaceutical Water-for-Injection service, or for use in explosive atmospheres. Applications of this kind require a different product line; contact Kestrel Fluid Systems before proceeding.

1.2 Safety symbols used in this manual

Symbol	Meaning
DANGER	Indicates a hazard which, if not avoided, will result in death or serious injury.
WARNING	Indicates a hazard which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a hazard which, if not avoided, may result in minor or moderate injury.
NOTICE	Indicates a practice which, if not avoided, may result in equipment damage.

1.3 General safety precautions

The S-200 operates at pressures of up to 22 bar. Never loosen, disconnect or adjust any fitting, pressure vessel end cap or instrument connection while the unit is pressurised. Always isolate the feed supply, stop the high-pressure pump and vent the system to atmosphere through the manual drain valve before opening any part of the pressure envelope.

The high-pressure pump motor is supplied from a 400 V three-phase circuit. Only personnel qualified to work on low-voltage industrial installations may open the control cabinet. The main isolator must be locked out and tagged before the cabinet door is opened. A residual voltage of up to 60 V may remain on the variable-frequency drive DC bus for 5 minutes after isolation; wait at least this long before touching drive terminals.

Membrane preservation solution supplied in new and stored units contains sodium metabisulphite. Avoid skin and eye contact and flush the system thoroughly as described in Section 6 before placing the unit into service. The safety data sheet is supplied separately with the shipping documentation.

Cleaning chemicals used in the membrane cleaning procedure described in Section 10 are strongly acidic or strongly alkaline. Personal protective equipment consisting of chemical-resistant gloves, face shield and apron must be worn during all cleaning operations.

1.4 Personnel qualification

Mechanical installation may be carried out by a competent pipefitter familiar with industrial water treatment plant. Electrical installation must be carried out by a qualified electrician. Commissioning, membrane replacement and chemical cleaning must be carried out by a technician who has completed the Kestrel service familiarisation programme or by a Kestrel-authorised service partner.

2. System Description

2.1 Process overview

Feed water enters the unit through the feed isolation valve and passes first through a 5 micron cartridge pre-filter, which protects the high-pressure pump and the membrane elements from particulate matter. A low-pressure switch downstream of the pre-filter inhibits pump start if inlet pressure falls below 1.0 bar.

The high-pressure pump, a multistage vertical centrifugal unit driven by a variable-frequency drive, raises the feed pressure to the operating value required by the membranes. Pressurised feed enters the first of two pressure vessels, each containing three 4-inch spiral-wound thin-film composite membrane elements arranged in series.

Water passing through the membrane becomes permeate and is collected in the permeate manifold. Water rejected by the membrane, carrying the concentrated dissolved solids, leaves through the concentrate manifold. A proportion of the concentrate is returned to the pump suction through the recycle loop to maintain adequate cross-flow velocity across the membrane surface; the remainder passes to drain through the motorised concentrate control valve.

Permeate quality is monitored continuously by an inline conductivity sensor. If permeate conductivity exceeds the configured alarm setpoint, the unit diverts permeate to drain through the three-way divert valve until quality recovers.

2.2 Main components

Item	Component	Description
1	Feed isolation valve	Manual ball valve, DN25, stainless steel
2	Cartridge pre-filter	Single 20-inch housing, 5 micron polypropylene element
3	Low-pressure switch	Adjustable, factory set at 1.0 bar
4	High-pressure pump	Multistage vertical centrifugal, 4.0 kW
5	Variable-frequency drive	Mounted inside control cabinet
6	Pressure vessels	Two vessels, FRP, 4-inch, 300 psi rating
7	Membrane elements	Six elements, type KFS-4040-BW
8	Concentrate control valve	Motorised globe valve with position feedback
9	Permeate divert valve	Three-way solenoid valve, normally to drain
10	Conductivity sensor	Two-electrode, with integral temperature compensation
11	Flow meters	Two paddlewheel sensors, permeate and concentrate
12	Control cabinet	IP54, containing PLC, HMI, drive and terminals

2.3 Options not included as standard

The following items are available as factory-fitted or retrofit options and are not part of the standard S-200 supply. Each option is covered by its own supplementary documentation, supplied with the option itself.

Antiscalant dosing package, consisting of a diaphragm metering pump, 60 litre chemical tank and level switch. Refer to supplementary manual KFS-DOS-011.

Activated carbon pre-treatment vessel for dechlorination of chlorinated feed water. Refer to supplementary manual KFS-CARB-004.

Permeate break tank and repressurisation set. Refer to supplementary manual KFS-TNK-002.

3. Technical Specifications

3.1 Performance data

Performance figures are stated at the reference conditions defined below. Actual output varies with feed water temperature, feed water salinity and membrane age.

Parameter	Value	Unit
Nominal permeate output	2,400	litres per hour
Nominal recovery rate	60	%
Nominal salt rejection, new membranes	98.5	%
Minimum salt rejection over warranty period	96.0	%
Reference feed temperature	15	degrees C
Reference feed conductivity	800	microsiemens per cm
Reference operating pressure	12	bar
Maximum operating pressure	22	bar
Permeate outlet pressure	0.5 to 1.5	bar

Permeate output falls by approximately 3 percent for every 1 degree C that feed water temperature falls below the reference temperature of 15 degrees C. At a feed temperature of 8 degrees C, expected output is therefore approximately 1,900 litres per hour.

3.2 Feed water requirements

The unit must not be operated outside the following feed water limits. Operation outside these limits will shorten membrane life and is not covered by the warranty.

Parameter	Limit
Feed pressure at inlet	Minimum 1.0 bar, maximum 6.0 bar
Feed temperature	Minimum 5 degrees C, maximum 35 degrees C
Feed conductivity	Maximum 3,000 microsiemens per cm
Silt Density Index (SDI15)	Maximum 4.0
Turbidity	Maximum 1.0 NTU
Free chlorine	Maximum 0.1 mg per litre
Iron, total	Maximum 0.05 mg per litre
Manganese	Maximum 0.05 mg per litre
pH, continuous operation	3.0 to 10.0

Parameter	Limit
pH, during cleaning	2.0 to 11.5
Oils and hydrocarbons	Must be absent

3.3 Electrical data

Parameter	Value
Supply voltage	400 V, three phase, plus neutral and earth
Supply frequency	50 Hz
Total connected load	4.8 kW
Full load current	9.2 A
Recommended upstream protection	16 A type C circuit breaker
Required earth leakage protection	30 mA type B residual current device
Control voltage	24 V DC, generated internally
Enclosure protection rating	IP54

3.4 Physical data and connections

Parameter	Value
Overall length	1,850 mm
Overall width	760 mm
Overall height	1,620 mm
Dry weight	285 kg
Operating weight	340 kg
Feed connection	DN25 male threaded, BSP parallel
Permeate connection	DN20 male threaded, BSP parallel
Concentrate to drain connection	DN20 hose tail, 25 mm
Required drain capacity	Minimum 1,800 litres per hour
Ambient temperature range	Minimum 5 degrees C, maximum 40 degrees C
Maximum ambient relative humidity	80 percent, non-condensing
Sound pressure level at 1 metre	68 dB(A)

4. Delivery, Handling and Site Requirements

4.1 Receiving the unit

On delivery, inspect the shipping crate for external damage before signing the carrier's delivery note. Any visible damage must be recorded on the delivery note at the time of receipt; claims for transit damage cannot be accepted once the delivery note has been signed clean.

Remove the crate and check the contents against the packing list. A standard shipment comprises the skid-mounted unit, a spares and documentation box, two lifting eyebolts, and a preservation certificate stating the date the unit was filled with preservation solution.

4.2 Lifting and handling

The skid frame is fitted with fork pockets on both long sides. Use a forklift of at least 1,000 kg capacity with forks fully engaged across the full width of the skid. Do not lift the unit by the pressure vessels, the pump, the pipework or the control cabinet.

Where forklift access is not available, the two supplied lifting eyebolts may be fitted to the tapped points at the top corners of the frame. Use a spreader beam to keep sling angles below 45 degrees from vertical. The unit must be lifted upright at all times; laying the unit on its side will damage the membrane elements.

4.3 Site requirements

The unit must be installed indoors on a level, load-bearing floor capable of supporting the operating weight given in Section 3.4. The floor should be finished to fall towards a drain, as small quantities of water are released during filter changes and sampling.

Allow a minimum clearance of 1,000 mm at the front of the unit for operator access to the control panel, and 1,200 mm at the end of the pressure vessels for membrane removal. A minimum of 300 mm is required on the remaining sides for ventilation.

The room must be frost-free at all times. If there is any possibility of the ambient temperature falling below 5 degrees C, the unit must be drained and preserved as described in Section 12 before the onset of cold weather.

A drain point capable of accepting the flow rate given in Section 3.4 must be available within 3 metres of the unit. The drain must be open to atmosphere; a direct connection to a pressurised or sealed drain line is not permitted, as back-pressure on the concentrate line disturbs the recovery control.

5. Installation

5.1 Mechanical installation

Position the skid in its final location and check that it is level in both directions using a spirit level placed on the frame. If the floor is uneven, shim under the frame feet; do not rely on the pipework to take up misalignment. Once level, secure the frame to the floor through the four fixing holes using M12 anchors.

Connect the feed line to the feed connection. A manual isolation valve and a strainer must be fitted in the feed line upstream of the unit if these are not already present in the site pipework. The feed line should be sized to deliver the required flow at the minimum inlet pressure given in Section 3.2.

Connect the permeate line to the permeate connection. The permeate line must run continuously downwards or be vented at its high point; trapped air in the permeate line causes erratic conductivity readings during start-up.

Connect the concentrate line to the drain. The concentrate line must terminate above the flood level of the drain with a visible air gap of at least 50 mm to prevent back-siphonage.

All connections should be made using flexible hoses or expansion joints at the unit so that pipework stresses are not transmitted into the skid. Support all site pipework independently of the unit.

> *Do not use hemp or PTFE tape on the permeate connections. Use only the supplied EPDM sealing washers; thread sealants contaminate the permeate and produce a persistent total organic carbon reading.*

5.2 Electrical installation

Confirm that the site supply matches the electrical data given in Section 3.3 before connecting. Connecting the unit to a 230 V supply, or to a supply without a neutral conductor, will damage the control transformer.

Route the supply cable into the control cabinet through the gland plate at the base of the cabinet. Terminate the three phases at terminals L1, L2 and L3, the neutral at terminal N, and the protective earth at the earth bar. The recommended supply cable is 5-core 2.5 square millimetre.

After connection, energise the supply with the main isolator off and confirm phase rotation using a phase rotation meter at the incoming terminals. The pump must rotate in the direction indicated by the arrow on the pump housing. Incorrect rotation reduces output and, if sustained, damages the pump seal.

Volt-free contacts for remote common alarm and remote run status are available at terminals X2:1 to X2:6. Contact rating is 250 V AC, 2 A. A remote start-stop input is available at terminals X2:7 and X2:8; a link is fitted across these terminals at the factory and must remain in place if remote start-stop is not used.

6. Commissioning

Commissioning must be carried out in the sequence given below. Do not attempt to bring the unit to full operating pressure until the flush described in Section 6.2 has been completed.

6.1 Pre-start checks

Confirm that all transit packing has been removed, that the cartridge pre-filter element is fitted and the housing correctly closed, that all manual valves are in the positions marked on the commissioning label, and that the concentrate line discharges freely to drain.

Check the pump oil level through the sight glass. The level should be at the centre of the glass with the unit stationary. Top up if required with the grade stated on the pump nameplate.

6.2 Preservation flush

New units and units returning from storage are filled with sodium metabisulphite preservation solution which must be flushed to drain before the unit produces usable permeate.

Open the feed isolation valve and allow the unit to fill at feed pressure with the high-pressure pump stopped. Open the manual vent valves at the top of each pressure vessel until water flows free of air, then close them.

Select FLUSH mode on the control panel. The unit will run the pump at minimum speed with the concentrate valve fully open and the permeate divert valve to drain. Run in this condition for 30 minutes.

At the end of the flush period, take a sample from the permeate sample point and confirm that no sulphite remains, using the test strips supplied in the spares box. Repeat the flush in 15 minute increments until the test is negative.

6.3 Initial performance run

Select AUTO mode. The unit will ramp the pump to the pressure required to achieve the permeate flow setpoint. Allow the unit to run for 60 minutes to allow the membranes to compact and the readings to stabilise.

Record the following values on the commissioning record sheet supplied in the documentation box: feed pressure, operating pressure, permeate flow, concentrate flow, feed conductivity, permeate conductivity and feed temperature. These values form the reference baseline against which future performance is judged, and must be normalised to the reference conditions using the correction chart in Appendix B.

The commissioning record sheet must be returned to Kestrel Fluid Systems within 30 days of start-up. Failure to return the sheet limits warranty cover on the membrane elements as described in Section 13.

7. Operation

7.1 Operating modes

Mode	Behaviour
AUTO	Normal production. The unit starts and stops on demand from the permeate tank level signal or the remote start-stop input, and modulates pump speed to hold the permeate flow setpoint.
FLUSH	Pump runs at minimum speed with concentrate valve fully open and permeate diverted to drain. Used at commissioning and after chemical cleaning.
MANUAL	Pump speed and concentrate valve position are set directly by the operator. Protective interlocks remain active. Intended for fault-finding only.
STANDBY	Pump stopped, unit pressurised, all alarms active. The unit performs an automatic 90 second flush every 24 hours to prevent biological growth on the membranes.
OFF	Pump stopped, control power on, alarms suppressed. The automatic standby flush does not run in this mode.

7.2 Starting the unit

Confirm that the feed supply is available and that the permeate destination can accept flow. Turn the main isolator to the on position and wait for the HMI to complete its start-up sequence, which takes approximately 40 seconds.

Select AUTO from the mode menu. The unit will carry out a 60 second inlet flush before starting the high-pressure pump. During the ramp, permeate is diverted to drain until conductivity falls below the divert setpoint, typically within 2 to 3 minutes of start.

7.3 Routine monitoring

Operators should record the readings listed below at least once per shift and compare them with the commissioning baseline after normalisation. A change in normalised permeate flow of more than 10 percent, or a change in normalised salt rejection of more than 3 percent, indicates that cleaning or investigation is required.

Reading	Typical value	Action threshold
Feed pressure	2 to 4 bar	Below 1.0 bar: check supply and pre-filter
Differential pressure across pre-filter	0.2 to 0.5 bar	Above 1.0 bar: replace cartridge
Operating pressure	10 to 14 bar	Above 18 bar: clean membranes
Permeate flow	2,400 l/h	Below 2,100 l/h normalised: clean membranes
Concentrate flow	1,600 l/h	Below 1,300 l/h: check concentrate valve
Permeate conductivity	10 to 25 uS/cm	Above 50 uS/cm: investigate rejection

7.4 Stopping the unit

For short stops, select STANDBY. The unit remains pressurised and performs the automatic daily flush. For stops longer than 72 hours, select OFF and carry out a manual flush before shutdown. For stops longer than 14 days, the unit must be preserved as described in Section 12.

8. Control Panel and Alarm Codes

8.1 Alarm handling

Alarms are divided into warnings, which are displayed but do not stop the unit, and trips, which stop the high-pressure pump immediately. All alarms are latched and must be acknowledged at the HMI before the unit can be restarted. The last 200 alarm events are retained in the event log and can be exported to a USB memory stick through the diagnostics menu.

8.2 Alarm code list

Code	Description	Type	Likely cause
A01	Low feed pressure	Trip	Feed supply interrupted, feed valve closed, or pre-filter blocked
A02	High operating pressure	Trip	Fouled membranes, concentrate valve closed, or permeate line blocked
A03	Pump motor overload	Trip	Mechanical obstruction, bearing failure, or supply phase loss
A04	High permeate conductivity	Warning	Membrane damage, O-ring leak, or high feed salinity
A05	Low permeate flow	Warning	Fouled membranes or low feed temperature
A06	High concentrate flow	Warning	Concentrate valve stuck open or position feedback fault
A07	Conductivity sensor fault	Warning	Sensor cable damaged or sensor fouled
A08	Temperature sensor fault	Warning	Sensor open circuit or short circuit
A09	Drive communication fault	Trip	Loss of communication between PLC and variable-frequency drive
A10	Drive internal fault	Trip	Refer to the drive display for the drive's own fault code
A11	Emergency stop active	Trip	Emergency stop button pressed or safety circuit open
A12	Pre-filter differential high	Warning	Cartridge element requires replacement
A13	Permeate divert timeout	Trip	Permeate quality did not recover within 10 minutes of start
A14	Cleaning due	Warning	Normalised performance has crossed the cleaning threshold
A15	Service interval reached	Warning	Scheduled service hours elapsed since last service reset

Alarm A13 has a fixed timeout of 10 minutes which cannot be changed from the operator menu. If a longer recovery period is genuinely required because of an unusually long permeate line, the value can be altered at the engineer access level; contact Kestrel service for the access code.

9. Routine Maintenance

The intervals given below assume continuous operation on feed water within the limits of Section 3.2. Units operating on poorer quality feed water, or intermittently, may require more frequent attention.

9.1 Maintenance schedule

Interval	Task
Every shift	Record operating readings; check for leaks; confirm no active alarms
Weekly	Check pre-filter differential pressure; check pump oil level; verify drain flows freely
Monthly	Replace cartridge pre-filter element; clean conductivity sensor; check concentrate valve travel
Every 3 months	Calibrate conductivity sensor against a reference solution; verify flow meter readings; check all electrical terminals for tightness
Every 6 months	Chemical clean membranes if not already triggered by performance; inspect pressure vessel end caps and O-rings; test emergency stop circuit

Interval	Task
Every 12 months	Replace pump oil; replace all elastomeric seals in the pre-filter housing; carry out full performance test and record normalised results
Every 24 to 36 months	Replace membrane elements; expected element life under reference conditions is 3 years

9.2 Replacing the cartridge pre-filter

Stop the unit and select OFF. Close the feed isolation valve. Open the vent screw on top of the filter housing, then open the drain cock at the base of the housing and allow the housing to empty.

Unscrew the housing bowl using the strap wrench supplied. Remove and discard the used element. Clean the inside of the bowl and inspect the housing O-ring; replace the O-ring if it shows any sign of flattening, nicking or hardening.

Lubricate the O-ring with the silicone grease supplied and fit a new element, ensuring it is seated correctly on the lower spigot. Refit the bowl hand tight, then a further quarter turn with the strap wrench. Do not overtighten.

Close the drain cock, open the feed isolation valve slowly, and vent the housing until water flows free of air. Close the vent screw and return the unit to service.

> *Record the change on the maintenance log and reset the pre-filter hours counter in the maintenance menu, otherwise alarm A12 will re-appear at the next differential pressure excursion.*

9.3 Conductivity sensor cleaning and calibration

Isolate and depressurise the permeate line before removing the sensor. Wipe the electrodes with a lint-free cloth moistened with isopropyl alcohol. Do not use abrasive materials, which alter the cell constant.

For calibration, immerse the clean sensor in a 84 microsiemens per cm reference solution at 25 degrees C and allow 5 minutes for thermal equilibrium before entering the calibration routine from the maintenance menu. A two-point calibration using a second solution at 1,413 microsiemens per cm is recommended annually.

10. Membrane Cleaning and Replacement

10.1 When to clean

Cleaning is required when any of the following occurs, judged on values normalised to the commissioning baseline: normalised permeate flow has fallen by 10 percent or more; normalised salt passage has increased by 5 percent or more; or the differential pressure across a pressure vessel has increased by 15 percent or more.

Cleaning at the correct time is important. Membranes that are allowed to foul heavily before cleaning frequently cannot be restored to their original performance, and the fouling layer becomes progressively harder to remove.

10.2 Selecting the cleaning chemical

Fouling type	Indication	Cleaning solution
Mineral scale	Rising differential pressure and rising salt passage together	Citric acid at pH 2.5, or hydrochloric acid at pH 2.0
Biological fouling	Rising differential pressure, falling flow, slime on pre-filter	Alkaline detergent at pH 11.0 with 0.1 percent sodium lauryl sulphate
Organic fouling	Gradual flow loss with little differential pressure change	Alkaline detergent at pH 11.0

Fouling type	Indication	Cleaning solution
Colloidal fouling	Rapid differential pressure rise after SDI excursion	Alkaline detergent at pH 11.0 followed by acid clean

Where both scaling and biological fouling are suspected, always carry out the alkaline clean first. Performing the acid clean first can fix organic and biological material onto the membrane surface and make it permanently unremovable.

10.3 Cleaning procedure

Connect the cleaning skid to the cleaning connections marked CIP-IN and CIP-OUT. Prepare the cleaning solution in the cleaning tank using permeate water, not feed water, and confirm the pH before circulating.

Circulate the solution through the pressure vessels at low pressure, not exceeding 4 bar, for 30 minutes. Maintain the solution temperature between 30 and 35 degrees C. Do not exceed 35 degrees C, as higher temperatures at extreme pH will damage the membrane.

Soak for a further 60 minutes with the circulation pump stopped, then circulate again for 30 minutes. Drain the cleaning solution, then flush with permeate until the pH of the water leaving the vessels is within 0.5 pH units of the flush water.

Return the unit to service in FLUSH mode for 30 minutes with permeate to drain before selecting AUTO. Record the post-clean performance and compare with the pre-clean values.

10.4 Replacing membrane elements

Depressurise and drain the unit. Remove the end cap retaining rings and withdraw the end caps. Push the elements out of the vessel from the feed end using the plastic push rod supplied; never use a metal bar.

Note the orientation of each element before removal. Elements must be reinstalled with the brine seal facing the feed end of the vessel. An element fitted the wrong way round will bypass feed water around the element and produce poor rejection with normal flow.

Lubricate the brine seals and interconnector O-rings with glycerine only. Petroleum-based lubricants attack the seals and the membrane.

Always replace all six elements at the same time. Mixing new and aged elements in the same vessel causes uneven flux distribution and shortens the life of the new elements.

11. Troubleshooting

Symptom	Possible cause	Action
Unit will not start, no display	Supply isolated, control fuse blown, or main isolator off	Check supply, check fuse F1 in the control cabinet, check isolator position
Unit trips on A01 immediately after start	Feed valve closed or pre-filter blocked	Open feed valve, check pre-filter differential, replace cartridge
Unit trips on A02 shortly after start	Concentrate valve failed closed or permeate line blocked	Check concentrate valve position feedback, confirm permeate line open
Permeate flow low, pressure normal	Low feed temperature or fouled membranes	Normalise the reading for temperature before concluding; clean if still low
Permeate flow low, pressure high	Scaled or fouled membranes	Carry out chemical cleaning per Section 10

Symptom	Possible cause	Action
Permeate conductivity high, flow normal	O-ring leak on an interconnector or damaged element	Probe each vessel to locate the faulty element; replace element and O-rings
Permeate conductivity high, flow also high	Element fitted the wrong way round after service	Remove and refit elements with brine seal facing the feed end
Pump noisy, output fluctuating	Air in the suction line or cavitation from low feed pressure	Vent the suction line, confirm feed pressure at least 1.0 bar during operation
Erratic conductivity reading at start-up	Air trapped in the permeate line	Vent the permeate line high point; ensure permeate line falls continuously
Alarm A09 recurring	Loose drive communication cable or electrical interference	Check cable termination and shield earthing; separate signal cable from power cable
Water on the floor beneath the skid	Pre-filter housing seal or pump mechanical seal leaking	Identify the source; replace the housing O-ring or arrange pump seal replacement

12. Storage, Preservation and Decommissioning

12.1 Short-term shutdown

For shutdowns of up to 14 days, leave the unit in STANDBY mode so that the automatic daily flush continues to operate. If control power must be removed, flush the unit manually every 3 days for 15 minutes.

12.2 Long-term preservation

For shutdowns longer than 14 days, the membranes must be preserved. Flush the unit with permeate for 30 minutes, then circulate a solution of 1 percent sodium metabisulphite in permeate water through the pressure vessels for 15 minutes. Close all valves to seal the vessels full of solution.

The preservation solution must be replaced every 6 months, or sooner if the pH of the solution falls below 3.0. Check and record the pH monthly during preservation.

> *Preserved units must be stored between 5 and 35 degrees C and protected from direct sunlight. Freezing destroys the membrane elements and the damage is not repairable.*

12.3 Decommissioning and disposal

Before disposal, drain all liquids and neutralise any residual cleaning chemicals. Membrane elements are composite plastics and should be disposed of as industrial waste in accordance with local regulations; they are not recyclable through standard plastics streams.

The control cabinet contains electronic assemblies which fall under waste electrical and electronic equipment regulations in most jurisdictions and must be separated from the metal frame before scrapping.

13. Warranty

13.1 Scope of cover

Kestrel Fluid Systems GmbH warrants that the S-200 unit will be free from defects in materials and workmanship for a period of 24 months from the date of despatch, or 18 months from the date of commissioning, whichever expires first.

Membrane elements are warranted separately for a period of 12 months from the date of commissioning, and only where the commissioning record sheet described in Section 6.3 has been returned within 30 days of start-up. Where the record sheet has not been returned, membrane cover is limited to 90 days from despatch.

The warranty covers the repair or replacement of defective parts at the discretion of Kestrel Fluid Systems. It does not cover labour for on-site diagnosis, removal or refitting, nor any consequential loss including loss of production, loss of profit, or the cost of alternative water supply.

13.2 Exclusions

The warranty is void where the unit has been operated outside the feed water limits given in Section 3.2, where the unit has been modified without written authorisation, where non-original spare parts have been fitted, where the unit has been allowed to freeze, or where the scheduled maintenance in Section 9 has not been carried out and recorded.

Consumable items are excluded from the warranty in all cases. Consumables comprise cartridge filter elements, O-rings and seals, pump oil, and cleaning chemicals.

13.3 Making a claim

Warranty claims must be submitted in writing within 14 days of the fault being discovered, quoting the unit serial number from the nameplate on the control cabinet door, together with the operating log covering the 30 days preceding the fault. Claims submitted without the operating log will be assessed on the available evidence and may be rejected.

14. Spare Parts

14.1 Recommended stock

The items listed below should be held on site to avoid unnecessary downtime. Quantities are for a single unit operating continuously.

Part number	Description	Recommended quantity
KFS-CF-2005	Cartridge filter element, 20 inch, 5 micron	12
KFS-OR-H20	Filter housing O-ring	2
KFS-4040-BW	Membrane element, brackish water, 4 inch	6
KFS-BS-4040	Brine seal for 4 inch element	6
KFS-IC-4040	Interconnector with O-rings	4
KFS-EC-KIT	Pressure vessel end cap seal kit	2
KFS-CS-100	Conductivity sensor	1
KFS-FM-PW	Paddlewheel flow sensor	1
KFS-PS-LOW	Low pressure switch	1
KFS-CAL-84	Calibration solution, 84 uS/cm, 500 ml	2
KFS-CAL-1413	Calibration solution, 1413 uS/cm, 500 ml	1
KFS-GR-SIL	Silicone grease for O-rings, 50 g	1

When ordering, always quote the unit serial number in addition to the part number. Component specifications have changed between revisions of the unit and the serial number allows the correct variant to be supplied.

14.2 Ordering

Spare parts orders should be sent to the service department at the address on the back cover. Standard stock items are normally despatched within 3 working days. Membrane elements are held in stock in Basel and are normally despatched within 5 working days.

15. Service and Support

15.1 Contacting service

Technical support is available from the Kestrel service department between 08:00 and 17:00 Central European Time, Monday to Friday, excluding Swiss public holidays. When contacting service, please have available the unit serial number, the revision of this manual, the active alarm codes, and the most recent set of recorded operating readings.

Service department telephone: +41 61 555 0142
Service department email: service@kestrelfluid.example
Spare parts email: parts@kestrelfluid.example

15.2 Service agreements

Kestrel offers scheduled service agreements covering the 6-monthly and 12-monthly tasks listed in Section 9. Agreements are arranged through the regional sales office and are not covered further in this manual.

15.3 Document history

Revision	Date	Change
1.0	June 2022	First issue
2.0	November 2023	Added variable-frequency drive; revised Section 8 alarm list
3.0	February 2025	Revised membrane element type to KFS-4040-BW; updated performance data
3.1	August 2025	Corrected electrical data in Section 3.3; added alarm A15
3.2	March 2026	Revised cleaning chemical selection table; clarified warranty conditions in Section 13

Kestrel Fluid Systems GmbH reserves the right to alter specifications without notice. This manual is supplied for the guidance of purchasers and no liability is accepted for errors or omissions.